1 Simulation 2: inference for the single pair comparison, higher dimension

Now we consider $\mathbf{X} = (X_1, \ldots, X_{50})$ is a higher dimensional random variable. In particular, $X_i \sim \text{Unif}(-\sqrt{3}, \sqrt{3})$ is independently generated for all $i \in [50]$. We then set $\boldsymbol{\beta} = (1/\sqrt{50}, \ldots, 1/\sqrt{50})^\top \in \mathbb{R}^{50}$.

$$\theta_i^*(\mathbf{X}) = \sin(i \cdot \pi/8) \cdot \phi(\boldsymbol{\beta}^\top \mathbf{X}), \quad \phi(t) = \frac{\tanh(t)}{0.628}, \tag{1}$$

and $i_0 = 1, j_0 = 4$, $S = 3$, $\Omega = \{\mathbf{X} \mid \boldsymbol{\beta}^\top \mathbf{X} > -0.5\}$.

2 Previous materials

For $\mathbf{A}$, we define

$$|\mathbf{A}| = \#\{(i,j) \mid n \geq i > j \geq 1, A_{ij} = 1\}. \tag{2}$$

Some other notations: $\psi'(t) = \exp(t)/\{\exp(t)+1\}^2$. $\delta(i \mid \mathbf{A}) = \{j \mid j \in [n], A_{ij} = 1\}$ a.k.a., all items j that have been compared with item i.

Algorithm 1: The generation process of $\mathcal{D}_n$

Input: Sampling probability p, Number of items n, the number of comparisons L.
Output: $\mathcal{D}_n = \{(\mathbf{X}_{ijl}, Y_{ijl}) \mid \text{when } A_{ij} = 1, i > j, i \in [n] \text{ and } l = 1, 2, \ldots, L\}$

- Generating Erdos-Renyi graph:
  - Initialize $\mathbf{A} = [A_{ij}]_{(i,j)\in[n]^2}$ as a $n \times n$ matrix with all entries $= 0$.
  - For all $(i,j) \in \{n \geq i > j \geq 1\}$, we generate $A_{ij} = 1$ with probability p and $A_{ij} = 0$ with probability $1-p$ (Bernoulli distribution).
  - For all $(i,j) \in \{n \geq i > j \geq 1\}$, set $A_{ji} = A_{ij}$.
- Generating $(\mathbf{X}_{ijl})$: For all $(i,j) \in \{n \geq i > j \geq 1\}, A_{ij} = 1$ and $l = 1, \ldots, L = 10^4$ or 10^3, we iid generate $X_{ijl} \sim \mathcal{X}$.
- Generating (Y_{ijl}): For all $(i,j) \in \{n \geq i > j \geq 1\}, A_{ij} = 1$ and $l = 1, \ldots, L = 10^4$ or 10^3, we iid generate Y_{ijl} such that

$$Y_{ijl} = 1 \text{ with probability } \frac{\exp\{\theta_i^*(\mathbf{X}_{ijl})\}}{\exp\{\theta_i^*(\mathbf{X}_{ijl})\} + \exp\{\theta_j^*(\mathbf{X}_{ijl})\}}, \tag{3}$$

  and $Y_{ijl} = 0$ with probability $1 - \frac{\exp\{\theta_i^*(\mathbf{X}_{ijl})\}}{\exp\{\theta_i^*(\mathbf{X}_{ijl})\}+\exp\{\theta_j^*(\mathbf{X}_{ijl})\}}$.

Algorithm 2: Estimating $\hat{\theta}_i(\cdot)$ for $i = 1, \ldots, n$

Input: Data set $\mathcal{D}_n$.
Output: $\hat{\theta}_i(\mathbf{x})$ for $i = 1, \ldots, n$

- Let $\mathcal{F}_\beta$ be the class of ReLU functions parametrized appropriately (we can try different parameters β to see the performance).
- Run SGD to minimize:

$$
\begin{aligned}
- \sum_{(i,j):\, i>j \text{ and } A_{ij}=1} \Bigg[\sum_{l=1}^{L(\mathcal{D}_n)} & Y_{ijl} \log \Big[\psi\big\{\theta_i(\mathbf{X}_{ijl} \mid \beta_i) - \theta_j(\mathbf{X}_{ijl} \mid \beta_j)\big\}\Big] \\
& + (1 - Y_{ijl}) \log \Big[1 - \psi\big\{\theta_i(\mathbf{X}_{ijl} \mid \beta_i) - \theta_j(\mathbf{X}_{ijl} \mid \beta_j)\big\}\Big]\Bigg],
\end{aligned} \tag{4}
$$

  over all possible $\theta_1, \ldots, \theta_n \in \mathcal{F}_\beta$. The functions $\theta_1, \ldots, \theta_n \in \mathcal{F}_\beta$ that SGD convergences to, are our estimators $\hat{\theta}_1, \ldots, \hat{\theta}_n$, respectively.
- if ReLU is still hard, consider function class like $\{\theta(\mathbf{x}) = \beta\mathbf{x} + \gamma\}$.

Algorithm 3: Estimating $\hat{\pi}_i(\mathbf{x})$ for $i = 1, \ldots, n$

Input: $\mathbf{A}$ and $\hat{\theta}_i(\mathbf{x})$ for $i = 1, \ldots, n$
Output: $\hat{\pi}_i(\mathbf{x})$ for $i = 1, \ldots, n$

- Obtain matrix $\hat{\boldsymbol{\Xi}}(\mathbf{x}) = \left[A_{ij}\psi'\left\{\hat{\theta}_i(\mathbf{x}) - \hat{\theta}_j(\mathbf{x})\right\}\right]_{(i,j)\in[n]^2}$ and A_{ij} is the ijth entry of $\mathbf{A}$.
- Obtain matrix $\hat{\boldsymbol{\mathscr{L}}}(\mathbf{x}) = \operatorname{diag}\left\{\hat{\boldsymbol{\Xi}}(\mathbf{x})(\underbrace{1, \ldots, 1}_{n \text{ times}})^\top\right\} - \hat{\boldsymbol{\Xi}}(\mathbf{x})$.
- Obtain the $\hat{\boldsymbol{\mathscr{L}}}^\dagger(\mathbf{x})$, the Moore-Penrose inverse of $\hat{\boldsymbol{\mathscr{L}}}(\mathbf{x})$.
- Obtain $(\hat{\pi}_1(\mathbf{x}), \ldots, \hat{\pi}_n(\mathbf{x}))^\top = \mathbb{I}(\mathbf{x} \in \Omega)|\mathbf{A}|\hat{\boldsymbol{\mathscr{L}}}^\dagger(\mathbf{x} \mid \mathbf{A})(\mathbf{e}_{i_0} - \mathbf{e}_{j_0})$. $\mathbf{e}_i$ is the standard basis of $\mathbb{R}^n$.

$$
\text{Get } \hat{\boldsymbol{\mathscr{L}}}^\dagger(\mathbf{x} \mid \mathbf{A})(\mathbf{e}_{i_0} - \mathbf{e}_{j_0}) \Leftrightarrow \text{Get } \mathbf{b} \text{ such that } \begin{pmatrix} \hat{\boldsymbol{\mathscr{L}}}(\mathbf{x} \mid \mathbf{A}) \\ \mathbf{1}_n^\top \end{pmatrix} \mathbf{b} = \begin{pmatrix} \mathbf{e}_{i_0} - \mathbf{e}_{j_0} \\ \mathbf{0}_n^\top \end{pmatrix} \tag{5}
$$

Algorithm 4: The principal debiased estimator for $\mathcal{Q}_{i_0j_0}(\Omega)$ and its variance estimation

Input: Comparison graph $\mathbf{A}$, samples $\mathcal{D}_n^*$, number of cross-fittings S and domain Ω.

Output: $\hat{\mathcal{Q}}_{i_0j_0}(\Omega) = \sum_{s=1}^S \hat{\mathcal{Q}}_{i_0j_0}^{(s)}(\Omega)/S$ and its variance estimator $\hat{V}_{i_0j_0}(\Omega) = \sum_{s=1}^S \hat{V}_{i_0j_0}^{(s)}(\Omega)/S$.

- Split $[L] = \{1, \dots, L\}$ equally into S subsets, namely, $\mathcal{J}_n^{(1)}, \dots, \mathcal{J}_n^{(S)}$, such that $\mathcal{J}_n^{(1)} \cup \dots \cup \mathcal{J}_n^{(S)} = [L]$. We then split $\mathcal{D}_n^*$ into $\mathcal{D}_n^{(1)}, \dots, \mathcal{D}_n^{(S)}$ as $\mathcal{D}_n^{(s)} = \{(\mathbf{X}_{ijl}, Y_{ijl}) \mid n \geq i > j \geq 1, A_{ij} = 1, l \in \mathcal{J}_n^{(s)}\}$;

- For $s = 1, \dots, S$, input $\mathcal{D}_n^{(-s)} = \mathcal{D}_n^* \setminus \mathcal{D}_n^{(s)}$ as $\mathcal{D}_n$ in Algorithm 2 and the outputs of Algorithm 2 are denoted by $\hat{\theta}_1^{(s)}(\mathbf{x}), \dots, \hat{\theta}_n^{(s)}(\mathbf{x})$;

- For $s = 1, \dots, S$ and all (i, j) such that $A_{ij} = 1$ and $n \geq i > j \geq 1$ and all $l \in \mathcal{J}_n^{(s)}$, obtain
$$(\hat{\pi}_1^{(s)}(\mathbf{X}_{ijl}), \dots, \hat{\pi}_n^{(s)}(\mathbf{X}_{ijl}))$$
as the output of Algorithm 3 with $\mathbf{A} = \mathbf{A}$ and $(\hat{\theta}_1(\mathbf{x}), \dots, \hat{\theta}_n(\mathbf{x})) = (\hat{\theta}_1^{(s)}(\mathbf{X}_{ijl}), \dots, \hat{\theta}_n^{(s)}(\mathbf{X}_{ijl}))$.

- For each $s \in [S]$, obtain the test statistics $\hat{\mathcal{Q}}_{i_0j_0}^{(s)}(\Omega)$ as
$$\frac{1}{|\mathbf{A}|L/S} \sum_{(\mathbf{X}_{ijl}, y_{ijl}) \in \mathcal{D}_n^{(s)}} \Bigg[\mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \left\{ \hat{\theta}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{j_0}^{(s)}(\mathbf{X}_{ijl}) \right\} + \left\{ \hat{\pi}_i^{(s)}(\mathbf{X}_{ijl}) - \hat{\pi}_j^{(s)}(\mathbf{X}_{ijl}) \right\} \left[y_{ijl} - \psi \left\{ \hat{\theta}_i^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_j^{(s)}(\mathbf{X}_{ijl}) \right\} \right] \Bigg].$$

- For each $s \in [S]$, obtain the variance estimator $\hat{V}_{i_0j_0}^{(s)}(\Omega)$ as
$$\hat{\sigma}^{(s)}(\mathbf{A}) = \frac{1}{|\mathbf{A}|^2 L/S} \sum_{i>j, A_{ij}=1,\, l \in \mathcal{J}_n^{(s)}} \left[\hat{\pi}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\pi}_{j_0}^{(s)}(\mathbf{X}_{ijl}) \right].$$

$$\hat{V}_{i_0j_0}^{(s)}(\Omega) = \frac{1}{|\mathbf{A}|^2 L^2/S^2} \sum_{i>j, A_{ij}=1,\, l \in \mathcal{J}_n^{(s)}} \left\{ \mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \{ \hat{\theta}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{j_0}^{(s)}(\mathbf{X}_{ijl}) \} - \hat{\mathcal{Q}}_{i_0j_0}(\Omega) \right\}^2 + \frac{1}{L} \hat{\sigma}^{(s)}(\mathbf{A}). \tag{6}$$

Algorithm 5: Simulation implementation

- We will specify all parameters in each specific numerical experiment. We will try many different parameters and the record the corresponding results.
- Calculate
$$\mathcal{Q}_{i_0 j_0}(\Omega) = \frac{1}{10^5} \sum_{i=1,\ldots,10^5} \mathbb{I}(\mathbf{X}_i \in \Omega) \cdot \{\theta^*_{i_0}(\mathbf{X}_i) - \theta^*_{j_0}(\mathbf{X}_i)\}, \tag{7}$$
where $\mathbf{X}_i$ are iid generated from $\mathcal{X}$.

1 Obtain $\mathcal{D}^*_n$ from Algorithm 1.

2 Obtain $\hat{\mathcal{Q}}_{i_0 j_0}(\Omega)$ and $\hat{V}_{i_0 j_0}(\Omega)$ from Algorithm 4.

3 Obtain the 95% CI
$$\left(\hat{\mathcal{Q}}_{i_0 j_0}(\Omega) - \sqrt{\hat{V}_{i_0 j_0}(\Omega)} \cdot z_{.975}, \quad \hat{\mathcal{Q}}_{i_0 j_0}(\Omega) + \sqrt{\hat{V}_{i_0 j_0}(\Omega)} \cdot z_{.975}\right), \tag{8}$$
where $z_{.975} = 1.96$ is the 0.975 quantile of the standard Gaussian distribution.

- Repeat the above procedures 1–3, $T = 100$ times
- (Empirical coverage rate) Among the T times, record the empirical coverage rate as
$$\frac{\text{Number of times that the CI contains } \mathcal{Q}_{i_0 j_0}(\Omega)}{T}. \tag{9}$$
- (Estimation error) Among the T times, record $\hat{\mathcal{Q}}_{i_0 j_0}(\Omega) - \mathcal{Q}_{i_0 j_0}(\Omega)$ in all rounds, and we will use a violin plot to present the estimation error.
- (Average CI length) Among the T times, record $2\sqrt{\hat{V}_{i_0 j_0}(\Omega)} \cdot z_{.975}$ in all rounds, and we will use the average of them to show the average confidence interval length.
- (Estimation Error of the trivial estimator) In each round T, calculate the estimation error of the trivial estimator:
$$\begin{aligned} \hat{\mathcal{Q}}^{(s),\text{trivial}}_{i_0 j_0}(\Omega) &= \frac{1}{|\mathbf{A}|L/S} \sum_{(\mathbf{X}_{ijl},\, y_{ijl}) \,\in\, \mathcal{D}^{(s)}_n} \left[\mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \left\{\hat{\theta}^{(s)}_{i_0}(\mathbf{X}_{ijl}) - \hat{\theta}^{(s)}_{j_0}(\mathbf{X}_{ijl})\right\}\right] \\ \hat{\mathcal{Q}}^{\text{trivial}}_{i_0 j_0}(\Omega) &= S^{-1} \sum_{s=1}^{S} \hat{\mathcal{Q}}^{(s),\text{trivial}}_{i_0 j_0}(\Omega). \end{aligned} \tag{10}$$
We report the trivial estimator error $\hat{\mathcal{Q}}^{\text{trivial}}_{i_0 j_0}(\Omega) - \mathcal{Q}_{i_0 j_0}(\Omega)$.